Financial Dashboard Project Report

Introduction

This project represents my journey of building a **Financial Performance Dashboard** using Power BI, starting from raw data and transforming it into a meaningful, insight-driven visualization.

The goal was not just to create charts, but to **understand the business behind the data**, identify patterns, and present insights that could help in decision-making.

Business Requirements

The objective of this project was to design an interactive dashboard that can:

- Track overall **financial performance using key KPIs**
- Compare **actual revenue against budget targets**
- Analyze **profitability (Gross Profit & EBITDA)**
- Monitor **cash inflows and outflows**
- Evaluate **product/service performance**
- Identify delays in customer payments using **receivables aging**

Tools & Technologies Used

- **Power BI** – Dashboard development and visualization
- **DAX (Data Analysis Expressions)** – KPI and calculated measures
- **Power Query** – Data cleaning and transformation
- **Excel** – Data source

1. Data Understanding & Initial Exploration

The first step in building this project was to thoroughly understand the dataset and ensure its quality before performing any transformations or visualizations.

The dataset consisted of a single table named **"Data"**, which contained financial and operational metrics across different regions, products, and time periods.

1.1 Data Quality Check

A complete data profiling process was conducted to verify the reliability of the dataset. The results were as follows:

- **All columns were 100% valid**
- **No errors detected (0%)**
- **No missing or empty values (0%)**

This confirmed that the dataset was clean and ready for analysis without requiring any major preprocessing.

1.2 Column Analysis & Structure

Each column was carefully analyzed to understand its data type, uniqueness, and business relevance.

Time Dimension

- **Month**
    - 24 distinct values
    - Represents the time period of transactions
    - Data type: Date

Geographical Dimension

- **Region**
    - 4 distinct values: North, South, East, West
    - Data type: Text
    - Used to analyze regional performance

Product/Service Dimension

- **Product/Service**
    - 5 distinct categories:
        - Product A
        - Product B
        - Product C
        - Service X
        - Service Y
    - Data type: Text
    - Helps in comparing product vs service performance

Financial Metrics

- **Revenue**
    - 476 distinct values (472 unique)
    - Data type: Whole Number
- **COGS (Cost of Goods Sold)**
    - 480 distinct & unique values
    - Data type: Decimal
- **Gross Profit**
    - 480 distinct & unique values
    - Data type: Decimal
- **Opex (Operating Expenses)**
    - 463 distinct (446 unique)
    - Data type: Whole Number
- **EBITDA**
    - 480 distinct & unique values
    - Data type: Decimal

Cash Flow Metrics

- **Cash Inflows**
    - 464 distinct (456 unique)

    - Data type: Whole Number
- **Cash Outflows**
    - 468 distinct (456 unique)
    - Data type: Whole Number

Working Capital Metrics

- **Receivables Aging (Days)**
    - 61 distinct values
    - Data type: Whole Number
- **Payables Aging (Days)**
    - 61 distinct values
    - Data type: Whole Number

Budgeting Metrics

- **Revenue Budget**
    - 476 distinct (472 unique)
    - Data type: Whole Number
- **Budget Variance %**
    - 480 distinct & unique values
    - Data type: Decimal

1.3 Key Observations from Data Exploration

- The dataset is **highly structured and clean**, requiring minimal preprocessing.
- Financial metrics such as Revenue, COGS, and EBITDA are **highly granular**, allowing detailed analysis.
- Presence of **budget vs actual data** enables performance tracking.
- Working capital indicators (Receivables & Payables Aging) allow **cash flow efficiency analysis**.
- Data spans **multiple regions and product categories**, enabling multidimensional insights.

2. KPI Visualization

After creating the KPI measures, the next step was to integrate them into the dashboard using **Card visuals**, ensuring that key financial insights are immediately visible to users.

2.1 KPI Card Design & Placement

The KPI cards were strategically placed at the **top section of the dashboard**, following standard BI design principles where high-level metrics are shown first.

This allows users to quickly assess the **overall financial health** before diving into detailed analysis.

2.2 Implemented KPI Cards (Based on Dashboard Output)

From the final dashboard:

- **Total Revenue → 24M**
- **Gross Profit % → 0.5 (50%)**
- **EBITDA % → 0.2 (20%)**
- **Net Cash → 688K**

2.3 Interpretation of KPI Values

Total Revenue (24M)

- Indicates strong overall sales performance
- Serves as the primary indicator of business scale

Gross Profit % (50%)

- Shows that the business retains **half of its revenue after COGS**
- Reflects **efficient cost management**

EBITDA % (20%)

- Indicates operational profitability after expenses

- Suggests **moderate operational efficiency**

Net Cash (688K)

- Positive value indicates **healthy cash flow**
- Confirms that the business is **generating more cash than it spends**

2.4 Visual Design Enhancements

To improve readability and user experience, the following design elements were applied:

- **Large font size** for key numbers
- **Clear labels** for each KPI
- **Icons added** to represent financial meaning (Revenue, Profitability, Cash)
- **Consistent color theme** aligned with the dashboard design
- **Card spacing and alignment** for a clean, executive look

2.5 Business Value of KPI Section

This KPI section serves as:

- A **quick decision-making layer** for stakeholders
- A **summary of financial performance** at a glance
- A **starting point** for deeper analysis using charts below

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2.6 Key Insight

Even without exploring detailed visuals, a user can instantly conclude:

- The business is **profitable (positive margins)**
- It is **operationally stable (EBITDA positive)**
- It maintains a **positive cash position**

3. Slicers & Interactive Filtering

After building the KPI section, the next step was to enhance the dashboard's usability by adding **interactive slicers**. These slicers allow users to dynamically filter data and explore insights based on specific dimensions.

3.1 Purpose of Adding Slicers

Slicers were implemented to:

- Enable **dynamic filtering** across all visuals
- Allow users to focus on **specific segments**
- Improve **dashboard interactivity and user control**
- Provide a **customized analysis experience**

3.2 Slicers Implemented in the Dashboard

Based on the final dashboard design, three key slicers were added on the **left panel**:

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1. Product/Service Slicer

Field Used:
Data[Product/Service]

Options Available:

- Product A
- Product B
- Product C
- Service X
- Service Y

Functionality:

- Filters all visuals based on the selected product or service
- Helps compare **product vs service performance**

🌍 2. Region Slicer

Field Used:
Data[Region]

Options Available:

- North
- South
- East
- West

Functionality:

- Enables regional analysis
- Helps identify **top-performing and underperforming regions**

📅 3. Timeline Slicer

Field Used:
Data[Month]

Type: Dropdown slicer

Functionality:

- Allows users to filter data by specific months
- Enables **time-based trend analysis**
- Default option set to **"All"** for full overview

3.3 Slicer Design & Layout

To maintain a clean and professional layout:

- All slicers were placed in a **vertical panel on the left side**
- Used a **button-style layout** for Product/Service and Region
- Applied a **consistent color theme** matching the dashboard
- Ensured proper spacing for **easy selection and readability**

3.4 Interaction Behavior

All slicers were configured to:

- **Cross-filter all visuals** on the dashboard
- Update:

- KPI Cards
- Line Charts
- Bar Charts
- Waterfall Chart
- Variance Analysis

This ensures a **fully connected and responsive dashboard experience**.

3.5 Key Benefits

- Users can instantly answer questions like:
  - "Which region is most profitable?"
  - "How does Product C perform over time?"
  - "What is the cash position for a specific month?"
- Eliminates the need for multiple reports by enabling **all-in-one analysis**

3.6 Insight

The addition of slicers transforms the dashboard from a **static report** into an **interactive decision-making tool**, allowing stakeholders to explore data at different levels without technical knowledge.

4. Revenue & Profit Trend Analysis

After implementing KPI cards and slicers, the next step was to build a **time-based trend analysis** to understand how the business performs across different months.

This was achieved using a **Line Chart** that compares Revenue and Profit over time.

4.1 Objective of the Chart

The purpose of this visualization is to:

- Track **monthly performance trends**
- Compare **Revenue vs Profit behavior**
- Identify **growth patterns and fluctuations**
- Detect **high-performing and low-performing months**

4.2 Chart Type Selection

- **Visualization Used:** Line Chart
- **Reason for Selection:**
  Line charts are best suited for **time-series analysis**, making it easy to observe trends, peaks, and drops over a period.

4.3 Measures Used

Two key measures were used to construct the chart:

Total Revenue

Total Revenue = SUM(Data[Revenue])

Total Gross Profit (Selected for Clarity)

Total Gross Profit = SUM(Data[Gross Profit])

Note: EBITDA could also be used, but Gross Profit was selected for clearer comparison with Revenue.

4.4 Chart Configuration in Power BI

The chart was built using the following field placements:

- **X-Axis → Month**
- **Y-Axis →**
    - Total Revenue
    - Total Gross Profit

This resulted in a **dual-line chart**, where:

- One line represents **Revenue**
- One line represents **Profit**

4.5 Visual Interpretation (Based on Dashboard)

From the chart:

- Revenue and Profit show **similar movement patterns**, indicating consistent cost structure
- There are **noticeable dips** around certain months (e.g., mid-year and October)
- Strong peaks observed in months like **June, September, and November**
- Profit line always stays below Revenue, as expected

4.6 Key Insights

- The business demonstrates **seasonal fluctuations** in performance
- Profit margins appear **relatively stable**, as both lines move together
- Sudden drops may indicate:
    - Increased costs
    - Lower sales volume
- Peak months can be targeted for **future strategic planning**

4.7 Design Enhancements Applied

- **Distinct colors** used for Revenue and Profit lines
- **Markers enabled** for better visibility of data points
- **Legend added** for clarity
- Clean layout to match overall dashboard theme

4.8 Business Value

This chart allows stakeholders to:

- Quickly understand **performance trends over time**
- Identify **critical months requiring attention**
- Make **data-driven decisions** for revenue and cost optimization

5. Budget vs Actual Variance Analysis

After analyzing revenue and profit trends, the next step was to evaluate how actual performance compares against planned targets. This was done using a **Budget vs Actual Variance chart**, which helps in measuring business performance against expectations.

5.1 Objective of the Chart

The primary goal of this visualization is to:

- Compare **Actual Revenue vs Budgeted Revenue**
- Identify **overperformance and underperformance**
- Track **variance trends across months**
- Support **financial planning and forecasting decisions**

5.2 Measures Used

To build this chart, the following measures were used:

Total Revenue (Actual)

```
Total Revenue = SUM(Data[Revenue])
```

Total Budget

```
Total Budget = SUM(Data[Revenue Budget])
```

Variance %

```
Variance % =
DIVIDE(
   SUM(Data[Revenue]) - SUM(Data[Revenue Budget]),
   SUM(Data[Revenue Budget]),
   0
)
```

Why DIVIDE was used:

- Prevents division errors when the budget is zero

- Ensures stable calculation

Formatting Applied:

- Converted into **Percentage (%)**

5.3 Chart Type Selection

- **Visualization Used:** Line and Clustered Column Chart
- **Reason:**
  - Columns → Compare Actual vs Budget
  - Line → Show Variance % trend

5.4 Chart Configuration in Power BI

Fields were assigned as follows:

- **X-Axis → Month**
- **Column Values →**
  - Total Revenue
  - Total Budget
- **Line Values →**
  - Variance %

5.5 Visual Interpretation (Based on Dashboard)

From the chart:

- Revenue and Budget are closely aligned in several months
- Variance % fluctuates between **positive and negative values**
- Some months show **negative variance**, indicating underperformance
- Peaks in variance indicate **significant deviation from targets**

5.6 Key Insights

- The business does not consistently meet its budget targets
- There are **periods of strong overperformance**, suggesting growth opportunities
- Negative variance months may indicate:
  - Poor sales performance

    - Overestimated budgets
- The fluctuation suggests a need for **better forecasting accuracy**

5.7 Design Enhancements Applied

- **Clustered columns** for clear comparison (Actual vs Budget)
- **Line overlay** for variance % trend
- **Dual-axis used** for better readability
- Color coding:
    - Revenue → Blue
    - Budget → Dark Blue
    - Variance → Red

5.8 Business Value

This visualization enables stakeholders to:

- Monitor **financial discipline and planning accuracy**
- Quickly identify **problematic months**
- Improve **budgeting strategies and forecasting models**
- Take corrective actions based on variance trends

5.9 Key Insight

The chart highlights that while revenue is strong, **consistency against budget is a challenge**, indicating a gap in planning vs execution alignment.

6. Product/Service Performance Analysis

To understand how different offerings contribute to overall business performance, a **Product/Service Performance analysis** was implemented. This visualization compares revenue and profit across all products and services.

6.1 Objective of the Chart

The purpose of this chart is to:

- Compare **performance across products and services**
- Identify **top-performing and low-performing offerings**
- Analyze **profitability contribution**
- Support **product-level decision making**

6.2 Measures Used

The following measures were used:

Total Revenue

Total Revenue = SUM(Data[Revenue])

Total Gross Profit

Total Gross Profit = SUM(Data[Gross Profit])

6.3 Chart Type Selection

- **Visualization Used:** Clustered Bar Chart

  Reason for Selection:

- Ideal for comparing multiple categories
- Provides clear horizontal comparison
- Easy to identify ranking among products/services

6.4 Chart Configuration in Power BI

Based on your implementation:

- **Y-Axis → Product/Service**
- **X-Axis →**
  - Total Revenue
  - Total Gross Profit

This results in **side-by-side bars** for each product/service.

6.5 Visual Interpretation (From Dashboard)

From the chart:

- **Product C** generates the highest revenue (~4.9M)
- **Service X** and **Product B** also perform strongly
- **Product A** shows the lowest performance (~4.5M revenue)
- Gross Profit follows a similar trend but at lower values

6.6 Key Insights

- Product C is the **top-performing offering**
- Revenue distribution across products is **relatively balanced**, with slight variations
- Profit trends closely follow revenue, indicating **consistent cost structure**
- Lower-performing products may need:
  - Better marketing
  - Pricing optimization
  - Cost control

6.7 Design Enhancements Applied

- Horizontal bars for better readability
- Consistent color scheme:
  - Revenue → Light Blue
  - Profit → Dark Blue
- Data labels added (e.g., 4.9M, 2.3M)
- Sorted for better comparison

6.8 Business Value

This chart enables stakeholders to:

- Identify **high-value products/services**
- Allocate resources more effectively
- Improve **product strategy and pricing decisions**
- Focus on **profit-driving segments**

6.9 Key Insight

The business shows **strong but slightly uneven performance across offerings**, with Product C leading and Product A lagging, indicating opportunities for optimization.

7. Cash Flow Waterfall Analysis

To provide a clear understanding of how cash moves within the business, a **Cash Flow Waterfall Chart** was implemented. This visualization explains how the company transitions from inflows to final net cash.

7.1 Objective of the Chart

The purpose of this chart is to:

- Show **cash movement step-by-step**
- Identify **sources and uses of cash**
- Explain how **Net Cash is derived**
- Present a **financial story visually**

7.2 Business Logic

The waterfall follows this structure:

Cash Inflows → Cash Outflows → Net Cash

This helps stakeholders understand:

- How much cash is generated
- How much is spent
- What remains as final cash position

7.3 Measures Created

Total Inflows

Total Inflows = SUM(Data[Cash Inflows])

Total Outflows

Total Outflows = SUM(Data[Cash Outflows])

Net Cash

Net Cash = [Total Inflows] - [Total Outflows]

7.4 Challenge Faced (Important Learning)

While building the waterfall chart, a key issue was encountered:

Problem:
Power BI waterfall charts **do not work directly with multiple measures** like normal charts.

- Initially, trying to add separate measures (Inflows, Outflows, Net Cash) **did not produce the correct waterfall structure**
- The chart required a **category-based approach instead of multiple fields**

7.5 Solution: Creating a Helper Table

To solve this, a **custom helper table** was created:

```
Cash Flow Steps =
DATATABLE(
   "Step", STRING,
   {
      {"Cash Inflows"},
      {"Cash Outflows"},
      {"Net Cash"}
   }
)
```

This table acts as the **category axis** for the waterfall chart.

7.6 Creating Dynamic Waterfall Measure

A dynamic measure was created using SWITCH:

```
Cash Flow Value =
SWITCH(
   SELECTEDVALUE('Cash Flow Steps'[Step]),
   "Cash Inflows", [Total Inflows],
   "Cash Outflows", -[Total Outflows],
   "Net Cash", [Net Cash]
)
```

Important Logic:

- Outflows are converted to **negative values**
- This ensures correct downward movement in the chart

7.7 Chart Configuration in Power BI

- **Chart Type → Waterfall Chart**

- **Category →**
    - Cash Flow Steps[Step]
- **Y-Axis →**
    - Cash Flow Value

7.8 Visual Interpretation (From Dashboard)

From the chart:

- **Cash Inflows → ~2.9M (Increase)**
- **Cash Outflows → ~-2.2M (Decrease)**
- **Net Cash → ~0.69M (Final Total)**

7.9 Key Insights

- The business generates **strong inflows**
- A significant portion is consumed by **outflows**
- Final Net Cash remains **positive**, indicating stability
- Cash management appears **controlled but can be optimized**

7.10 Design Enhancements Applied

- Blue color for **increases (Inflows)**
- Red color for **decreases (Outflows)**
- Final total highlighted separately
- Data labels added for clarity

7.11 Business Value

This chart allows stakeholders to:

- Understand **cash movement at a glance**
- Identify **major cost impact (outflows)**
- Evaluate **financial stability**
- Make decisions on **cost control and cash optimization**

7.12 Key Insight

👉 Even though inflows are strong, **outflows consume a large portion**, making cash efficiency a key area for improvement.

8. Receivables Aging Analysis (Payment Behavior)

To evaluate customer payment patterns and working capital efficiency, a **Receivables Aging Analysis** was implemented. This chart highlights how long receivables remain outstanding across different regions.

8.1 Objective of the Chart

The purpose of this visualization is to:

- Analyze **customer payment behavior**
- Identify **delayed payments**
- Compare **receivables across regions**
- Improve **cash flow management**

8.2 Data Used

The chart is based on:

- **Receivables Aging (Days)** → grouped into aging buckets
- **Region** → East, North, South, West
- **Receivables Value** → aggregated as Total Receivables

8.3 Chart Type Selection

- **Visualization Used:** Clustered Column Chart

Reason:

- Allows comparison across **multiple regions**
- Clearly shows differences in **aging buckets**
- Easy to identify patterns in payment delays

8.4 Chart Configuration in Power BI

Based on your implementation:

- **X-Axis → Aging Bucket**
- **Y-Axis → Total Receivables**

- **Legend → Region**

8.5 Visual Interpretation (From Dashboard)

From the chart:

- East region shows the **highest receivables in early buckets (~3.4M)**
- West region has relatively **lower early receivables but higher in later buckets**
- North and South show **moderate distribution across buckets**
- Some regions show **increasing receivables with aging**, indicating delays

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8.6 Key Insights

- A significant portion of receivables lies in **short-term buckets**, which is healthy
- However, presence in higher aging buckets indicates:
    - **Delayed collections**
    - Potential **credit risk**
- Regional differences suggest:
    - Some regions have **better payment discipline**
    - Others may need **stricter credit control**

8.7 Design Enhancements Applied

- Distinct colors for each region
- Grouped columns for easy comparison
- Data labels for clarity
- Clear title: *Receivables Aging by Region (Payment Behavior)*

8.8 Business Value

This chart helps stakeholders to:

- Monitor **collection efficiency**
- Identify **high-risk regions**
- Improve **credit policies**
- Optimize **working capital management**

8.9 Key Insight

While most receivables are in early stages, **delayed payments in certain regions indicate potential cash flow risks**.

9. Challenges Faced

During the project, several challenges were encountered:

1. Waterfall Chart Issue

- Power BI does not support multiple measures directly
- Required creating a **helper table + SWITCH logic**

2. Variance Calculation

- Ensuring correct % calculation using DIVIDE
- Avoiding division errors

3. Data Understanding

- Identifying relationships between:
  - Revenue
  - Profit
  - Cash Flow

4. Visual Design

- Maintaining **clean layout and alignment**
- Choosing correct chart types for each analysis

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10. Recommendations for Improvement

- **Improve budget planning by using past trends and seasonality**
- **Focus more on high-profit products/services**
- **Reduce unnecessary costs to improve cash flow**
- **Analyze low-performing months and take corrective actions**
- **Improve collection process to reduce payment delays**
- **Create region-wise strategies to improve performance**
- **Regularly track variance to control performance gaps**

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11. Conclusion

This project successfully demonstrates the development of a **comprehensive Financial Dashboard using Power BI**, integrating:

- KPI tracking
- Trend analysis
- Budget vs actual comparison
- Product performance evaluation
- Cash flow visualization
- Receivables analysis